Cosmological Critical Coherence Index with Growth Connectivity and Robustness Margins

A Multi-Sector Structural-Stress Benchmark Using $H(z)$ and $f\sigma_8(z)$

Christof Krieg
MAAT Project – Independent Research, Germany
maat-research.com

2026

Abstract

The first cosmological Critical Coherence Index introduced a reduced structural-stress observable based on expansion rate, redshift activity, linear growth coherence, and structural imbalance. However, its connectivity and robustness sectors were specified only as operational targets, and the sector weights and transition cutoffs were fixed by convention.

This paper closes that operational gap at benchmark level. We extend the cosmological CCI to a multi-sector diagnostic using both Cosmic Chronometer $H(z)$ measurements and BOSS DR12 $f\sigma_8(z)$ growth data. The growth data define an explicit connectivity support $V_{\mathrm{corr}}(z)$, while joint expansion–growth consistency defines a robustness support $R_{\mathrm{robust}}(z)$. We also infer companion sector weights λ_a by a small maximum-entropy matching problem and define a data-driven transition proxy z_c from the curvature of $\log \widehat{\mathrm{CCI}}_{\cos}(z)$.

In the benchmark implementation, the inferred transition proxy is $z_c \simeq 1.114$, and the fitted companion weights follow the hierarchy $\lambda_S > \lambda_H > \lambda_R > \lambda_V$. The normalised multi-sector CCI reaches approximately 6.60 at $z = 1$ and 15.42 at $z = 2$. These results should not be interpreted as precision cosmological inference. They show instead that the cosmological CCI can be made operationally measurable beyond the chronometer-only projection by adding growth connectivity, robustness margins, calibrated sector weights, and a data-driven transition criterion.

1 Status and Scope

This work is a diagnostic benchmark, not a new cosmological model. It does not fit cosmological parameters, replace ΛCDM, or claim evidence for modified gravity. Its purpose is narrower:

> *Can the missing connectivity, robustness, weight, and cutoff pieces of the cosmological CCI be made operational using available expansion and growth data?*

The answer given here is yes, at the level of a reproducible benchmark. The construction turns the operational targets of the chronometer-only cosmological CCI into measurable proxy sectors:

$$H(z) \quad \text{for expansion consistency,} \qquad f\sigma_8(z) \quad \text{for growth connectivity,}$$

and a joint support margin for robustness. The result is a multi-sector structural-stress observable rather than a precision likelihood.

2 From v0.2 to v0.3

The previous cosmological CCI projection used the reduced form

$$\mathrm{CCI}_{\mathrm{v0.2}}(z) = \frac{E(z)(1+z)\,[1+U_{\mathrm{struct}}(z)]}{1+D(z)+\epsilon}, \tag{1}$$

where $E(z) = H(z)/H_0$ and $D(z)$ is a linear growth-factor proxy. This scalar expression is useful, but leaves three important questions open:

- How should connectivity V be measured?
- How should robustness R be measured?
- Can sector weights and transition cutoffs be inferred rather than fixed?

The present v0.3 benchmark answers these questions operationally. It uses $f\sigma_8(z)$ measurements to define a growth-connectivity sector, combines expansion and growth residuals into a robustness sector, calibrates companion weights λ_a, and estimates a transition proxy from the curvature of the resulting CCI curve.

3 Reference Cosmology and Data

The reference background is a flat Planck-2018 ΛCDM cosmology [1] with

$$H_0 = 67.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \qquad \Omega_m = 0.315, \qquad \Omega_\Lambda = 0.685, \qquad \sigma_{8,0} = 0.811. \tag{2}$$

The expansion function is

$$E(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}. \tag{3}$$

The growth factor is approximated by the Carroll–Press–Turner growth suppression formula [9]:

$$D(z) = \frac{g(z)}{g(0)(1+z)}, \tag{4}$$

with

$$g(z) = \frac{5\Omega_m(z)}{2}\left[\Omega_m(z)^{4/7} - \Omega_\Lambda(z) + \left(1 + \frac{\Omega_m(z)}{2}\right)\left(1 + \frac{\Omega_\Lambda(z)}{70}\right)\right]^{-1}. \tag{5}$$

The growth rate is approximated by the standard growth-index form [10]

$$f(z) = \Omega_m(z)^\gamma, \qquad \gamma = 0.55, \tag{6}$$

so that the reference growth observable is

$$f\sigma_8^{\mathrm{ref}}(z) = f(z)\,\sigma_{8,0} D(z). \tag{7}$$

The expansion sector uses the same 31-point Cosmic Chronometer compilation used in the chronometer-only projection [2, 3, 4, 5, 6, 7, 8]. The growth-connectivity sector uses BOSS DR12 consensus $f\sigma_8$ measurements [11]. The benchmark includes only a compact three-point growth table, so the results should be read as a proof of operational closure rather than as a full redshift-space-distortion likelihood.

4 Operational Sector Definitions

All sectors are converted into nonnegative defects $d_a(z)$ and dimensionless supports

$$\Gamma_a(z) = \frac{1}{1 + d_a(z)}. \tag{8}$$

This convention makes every support lie in $(0, 1]$, where larger values represent better consistency.

4.1 Expansion Consistency

For each chronometer point, define the expansion pull

$$p_H(z_i) = \frac{H_{\rm obs}(z_i) - H_{\Lambda\rm CDM}(z_i)}{\sigma_H(z_i)}. \tag{9}$$

The local expansion defect is a Gaussian-kernel average,

$$d_H(z) = \frac{\sum_i \exp\left[-\frac{(z-z_i)^2}{2\sigma_H^{2,\rm ker}}\right] p_H(z_i)^2}{\sum_i \exp\left[-\frac{(z-z_i)^2}{2\sigma_H^{2,\rm ker}}\right]}, \tag{10}$$

with $\sigma_H^{\rm ker} = 0.12$. The corresponding support is

$$\Gamma_H(z) = \frac{1}{1 + d_H(z)}. \tag{11}$$

4.2 Activity / Structural Imbalance

The activity-like sector is kept from the reduced CCI as a structural imbalance between expansion stress and linear growth coherence:

$$d_S(z) = U_{\rm struct}(z) = \frac{|E(z) - D(z)|}{E(z) + D(z) + \epsilon}. \tag{12}$$

This is not a thermodynamic entropy production rate. It is a dimensionless cosmological activity proxy measuring how strongly the expansion sector dominates the linear coherence proxy.

4.3 Growth Connectivity

The connectivity sector is defined from $f\sigma_8$ residuals. For each growth-data point,

$$p_V(z_j) = \frac{f\sigma_{8,\rm obs}(z_j) - f\sigma_{8,\rm ref}(z_j)}{\sigma_{f\sigma_8}(z_j)}. \tag{13}$$

The local growth-connectivity defect is

$$d_V(z) = \frac{\sum_j \exp\left[-\frac{(z-z_j)^2}{2\sigma_V^{2,\rm ker}}\right] p_V(z_j)^2}{\sum_j \exp\left[-\frac{(z-z_j)^2}{2\sigma_V^{2,\rm ker}}\right]}, \tag{14}$$

with $\sigma_V^{\rm ker} = 0.08$. The measured connectivity support is then

$$V_{\rm corr}(z) = \Gamma_V(z) = \frac{1}{1 + d_V(z)}. \tag{15}$$

4.4 Robustness Margin

Robustness is defined as joint consistency of expansion and growth:

$$d_R(z) = \frac{1}{2}\left[d_H(z) + d_V(z)\right], \qquad R_{\rm robust}(z) = \Gamma_R(z) = \frac{1}{1 + d_R(z)}. \tag{16}$$

This is deliberately simple. It measures whether the same redshift region is simultaneously supported by both expansion and growth data. In future precision applications, additional robustness terms should include covariance, stability priors, consistency between multiple surveys, and theoretical viability conditions.

5 Multi-Sector Cosmological CCI

The v0.3 cosmological CCI is defined as

$$\boxed{\mathrm{CCI}_{\mathrm{v0.3}}(z) = \frac{E(z)(1+z)\,[1+U_{\mathrm{struct}}(z)]}{1+D(z)+V_{\mathrm{corr}}(z)+R_{\mathrm{robust}}(z)+\epsilon}} \tag{17}$$

and the plotted quantity is normalised by its lowest-redshift grid value,

$$\widehat{\mathrm{CCI}}_{\mathrm{v0.3}}(z) = \frac{\mathrm{CCI}_{\mathrm{v0.3}}(z)}{\mathrm{CCI}_{\mathrm{v0.3}}(z_{\mathrm{min}})}. \tag{18}$$

Equation (17) should be read as an operational benchmark formula, not as the final cosmological MAAT functional. Its significance is that V and R are no longer merely named sectors: they are measured from growth and joint consistency data.

6 Companion Maximum-Entropy Weight Calibration

To test whether a multi-sector structural score can be calibrated from the same defect fields, we define the companion functional

$$\mathcal{F}_{\cos}(z) = \lambda_H d_H(z) + \lambda_S d_S(z) + \lambda_V d_V(z) + \lambda_R d_R(z). \tag{19}$$

This linear form is used only for weight calibration. It is monotonically related to the logarithmic support form used in the broader structural-selection framework,

$$-\sum_a \lambda_a \log(\epsilon + \Gamma_a)\,, \tag{20}$$

in the small-defect or chosen-cost sense. The linear proxy makes the calibration transparent while preserving the defect-ranking interpretation.

The weights are fitted by a small maximum-entropy matching problem. Let $\mathcal{B}$ be the best-fraction set selected by the unweighted defect sum, using the lowest 20% of grid points. The target vector is

$$\bar{d}_a^{\mathcal{B}} = \frac{1}{|\mathcal{B}|}\sum_{z_i\in\mathcal{B}} d_a(z_i). \tag{21}$$

The fitted weights minimise the mismatch between the ensemble expectation under

$$P_\lambda(z_i) = \frac{\exp[-\mathcal{F}_{\cos}(z_i)]}{\sum_k \exp[-\mathcal{F}_{\cos}(z_k)]} \tag{22}$$

and the target vector, with a weak prior and a weak share-balancing penalty:

$$\mathcal{L}(\lambda) = \sum_a \left[\langle d_a\rangle_\lambda - \bar{d}_a^{\mathcal{B}}\right]^2 + \eta_{\mathrm{prior}}\sum_a (\log\lambda_a)^2 + \eta_{\mathrm{bal}}\sum_a\left(\frac{\lambda_a}{\sum_b\lambda_b} - \frac{1}{4}\right)^2. \tag{23}$$

In the benchmark run,

$$\eta_{\mathrm{prior}} = 0.02, \qquad \eta_{\mathrm{bal}} = 0.01.$$

The prior keeps the fit from producing arbitrary extreme weights in a small dataset, while the balance term prevents one sector from dominating purely because of scale artefacts.

7 Data-Driven Transition Proxy

The chronometer-only paper used interpretive redshift bands rather than a data-derived transition. Here we introduce a simple operational cutoff proxy:

$$z_c = \arg\max_z \left| \frac{\mathrm{d}^2}{\mathrm{d}z^2} \log \widehat{\mathrm{CCI}}_{\mathrm{v0.3}}(z) \right| . \tag{24}$$

This definition locates the strongest curvature feature of the logarithmic structural-stress trajectory. It is not a physical phase transition by itself, but it provides a reproducible redshift marker for later comparison across models and datasets.

8 Results

8.1 Growth Connectivity

Figure 1 shows the growth input used to activate the connectivity sector. The BOSS DR12 $f\sigma_8$ points are compared with the Planck-ΛCDM growth-index proxy. The pull RMS in this compact growth table is 0.697, so the benchmark is consistent with the reference model at the level needed for a diagnostic test.

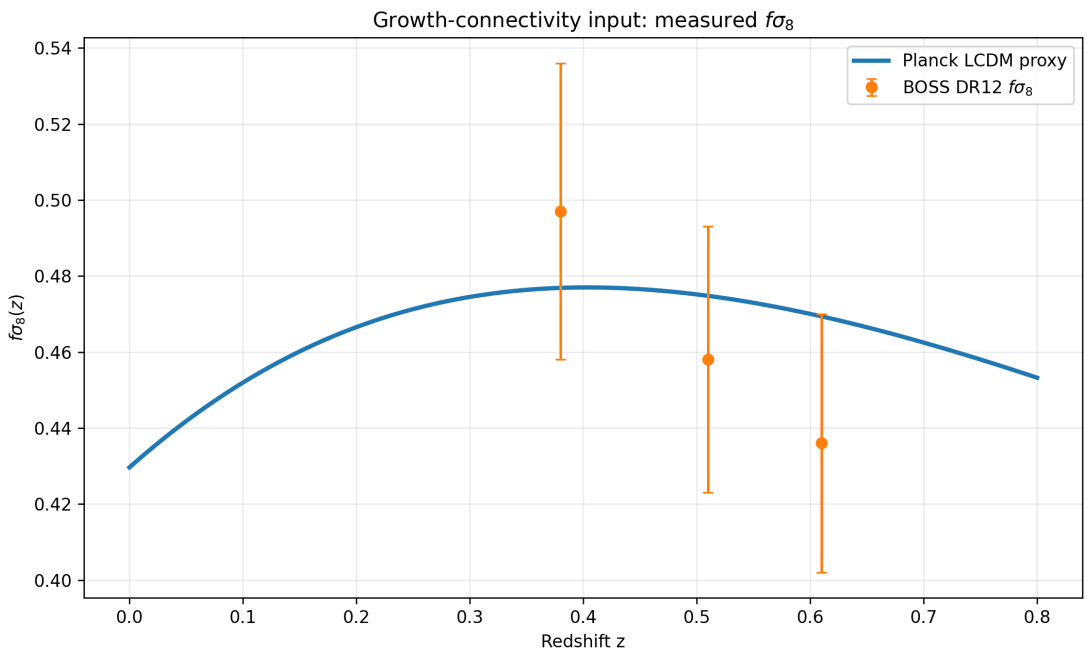

Figure 1: Growth-connectivity input. BOSS DR12 $f\sigma_8$ measurements are shown against the Planck-ΛCDM growth-index proxy. These residuals define the operational connectivity defect $d_V(z)$ and support $V_{\mathrm{corr}}(z)$.

8.2 Operational Supports

Figure 2 shows the expansion, connectivity, and robustness supports. The minimum support values in the grid are

$$\min V_{\mathrm{corr}} \simeq 0.510, \qquad \min R_{\mathrm{robust}} \simeq 0.460.$$

Thus the v0.3 construction does not merely add symbolic sectors: the additional supports vary nontrivially across redshift.

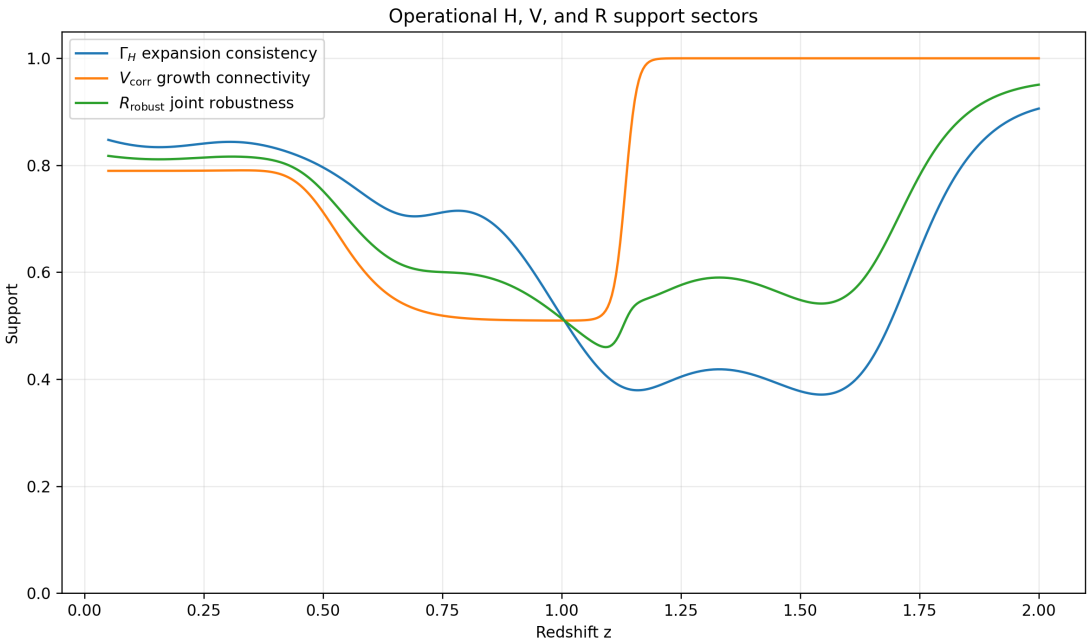

Figure 2: Operational support sectors in the v0.3 benchmark. Γ_H measures local expansion consistency, $V_{\rm corr}$ measures growth connectivity from $f\sigma_8$ residuals, and $R_{\rm robust}$ measures joint expansion–growth consistency.

8.3 CCI Comparison

Figure 3 compares the chronometer-only v0.2 projection with the multi-sector v0.3 projection. Because the denominator now includes both $V_{\rm corr}$ and $R_{\rm robust}$, the growth-supported diagnostic is less steep than the reduced v0.2 curve. The normalised v0.3 index reaches

$$\widehat{\rm CCI}_{\rm v0.3}(z=1) \simeq 6.60, \qquad \widehat{\rm CCI}_{\rm v0.3}(z=2) \simeq 15.42.$$

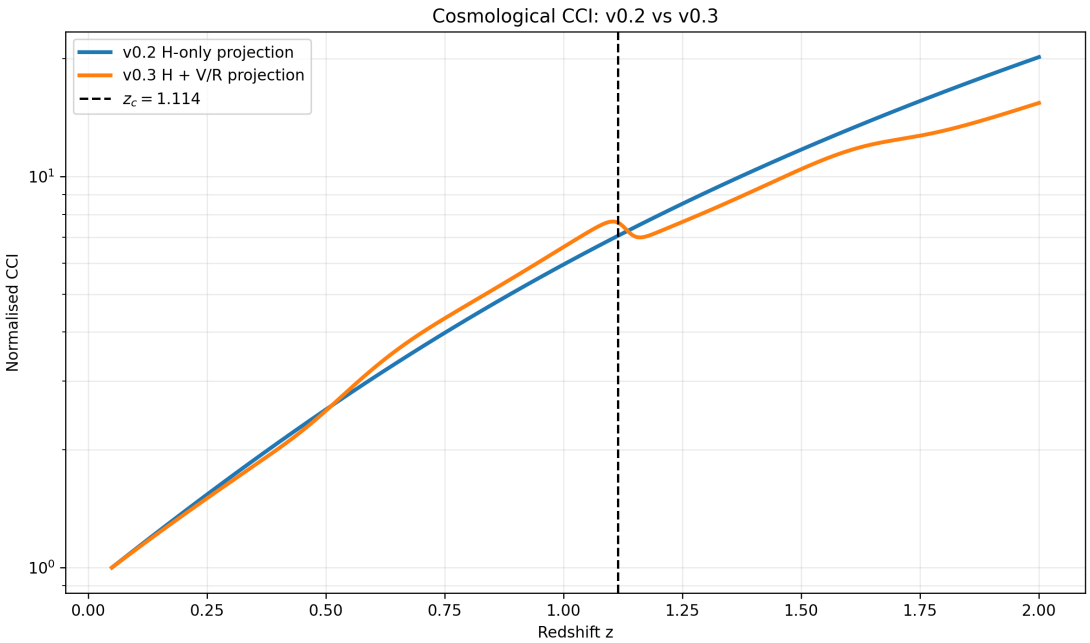

Figure 3: Comparison of the reduced v0.2 CCI and the multi-sector v0.3 CCI. The v0.3 curve includes measured growth connectivity and robustness margins. The dashed line marks the curvature-derived transition proxy $z_c \simeq 1.114$.

8.4 Transition Proxy

The curvature proxy of Eq. (24) yields

$$z_c = 1.1140311804.$$

Figure 4 displays the curvature curve used to define this marker. The value should not be interpreted as a sharp cosmic phase transition. It is a reproducible diagnostic marker for where the logarithmic multi-sector structural-stress trajectory changes most strongly in the tested redshift range.

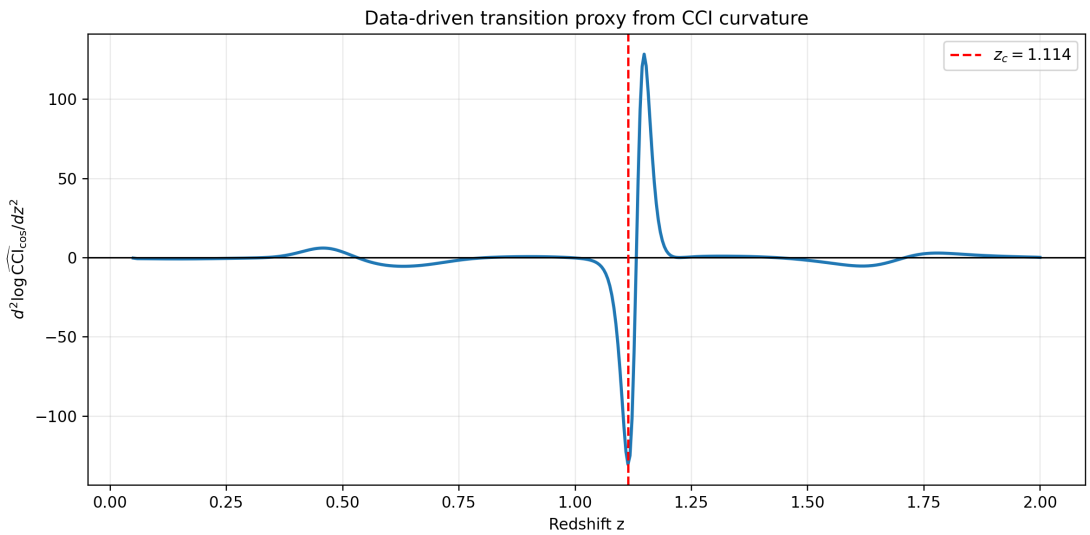

Figure 4: Curvature of $\log \widehat{\mathrm{CCI}}_{\mathrm{v0.3}}(z)$. The maximum absolute curvature within the interior grid defines the transition proxy $z_c \simeq 1.114$.

8.5 Calibrated Companion Weights

The maximum-entropy companion fit gives

$$\lambda_H = 1.7738789301, \tag{25}$$
$$\lambda_S = 1.9607028224, \tag{26}$$
$$\lambda_V = 1.0508708189, \tag{27}$$
$$\lambda_R = 1.2677133848. \tag{28}$$

The corresponding normalised shares are

$$H : 0.293, \qquad S : 0.324, \qquad V : 0.174, \qquad R : 0.209.$$

The hierarchy is therefore

$$\lambda_S > \lambda_H > \lambda_R > \lambda_V.$$

In words, the benchmark identifies activity/imbalance and expansion consistency as the strongest sectors in this small cosmological dataset, with robustness and connectivity still active but subdominant.

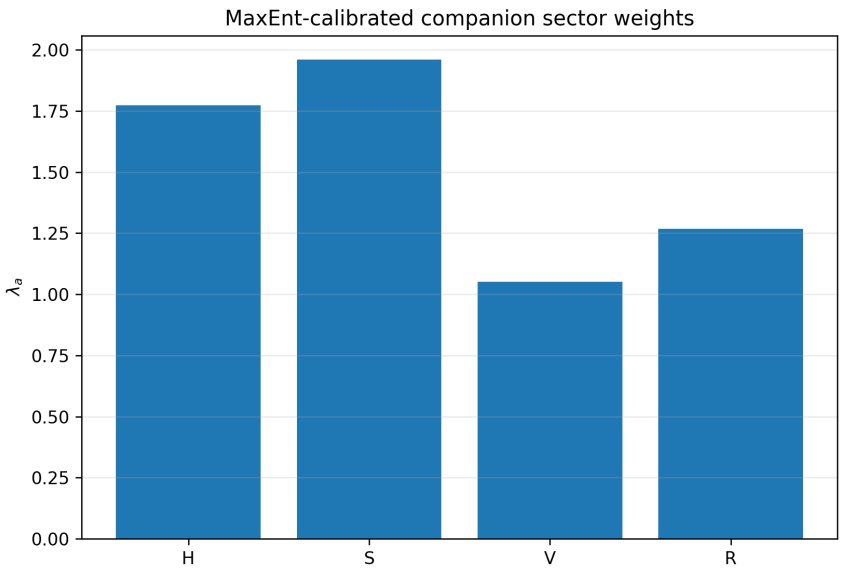

Figure 5: Maximum-entropy companion sector weights for the v0.3 benchmark. These are effective benchmark weights, not universal constants.

Table 1: Summary of numerical outputs for the v0.3 benchmark.

Quantity	Value
Chronometer points	31
BOSS DR12 $f\sigma_8$ points	3
$H(z)$ pull RMS	0.693
$f\sigma_8$ pull RMS	0.697
$\widehat{\mathrm{CCI}}_{\mathrm{v0.3}}(z=1)$	6.6006
$\widehat{\mathrm{CCI}}_{\mathrm{v0.3}}(z=2)$	15.4152
Transition proxy z_c	1.1140
$\min V_{\mathrm{corr}}$	0.5096
$\min R_{\mathrm{robust}}$	0.4599
Weight hierarchy	$\lambda_S > \lambda_H > \lambda_R > \lambda_V$

9 Interpretation

The key result is not that the universe has a new measured transition at $z \simeq 1.114$. The key result is operational:

> *The cosmological CCI can be extended from an expansion-growth scalar into a multi-sector observable with measured connectivity, robustness, calibrated companion weights, and a reproducible transition marker.*

This makes the framework more falsifiable. If future BAO, DESI, weak-lensing, or full redshift-space-distortion data produce inconsistent V_{corr} or R_{robust} sectors, the multi-sector CCI will expose the tension directly. Conversely, if different cosmological probes agree, the supports should remain high and the structural-stress curve should stabilise under dataset enlargement.

10 Limitations

The present benchmark has important limitations:

- the $f\sigma_8$ sector uses only three BOSS DR12 consensus points;
- no full covariance matrix is used;
- the growth factor is approximated rather than solved with a Boltzmann or numerical perturbation pipeline;
- the Gaussian kernel widths are operational choices, not fitted physical scales;
- the companion λ_a values are effective benchmark weights, not universal MAAT constants;
- the transition proxy z_c is a curvature marker, not a proven physical phase transition;
- BAO, supernovae, weak lensing, DESI, and model comparison are not included;
- the robustness support includes only joint expansion–growth residuals, not full theoretical stability conditions.

These limitations are intentional for this stage. The purpose is to close the previously missing operational definitions, not to perform final precision cosmology.

11 Reproducibility

The complete code and generated artifacts are archived in the public repository:

https://github.com/Chris4081/structural-selection-principle/tree/main/experiments/cosmological_cci_v03

The benchmark can be reproduced with:

```
cd experiments/cosmological_cci_v03
python3 maat_cci_cosmology_v03_growth.py
```

This regenerates the grid CSV, the result JSON, and all figures used in this paper:

- `maat_cci_cosmology_v03_grid.csv`,
- `maat_cci_cosmology_v03_results.json`,
- `plots/fsigma8_growth_connectivity.png`,

- `plots/v_r_supports.png`,
- `plots/cci_v02_v03_comparison.png`,
- `plots/transition_curvature.png`,
- `plots/lambda_calibration.png`.

The reproducibility loop is

$$\begin{aligned}&\text{Planck parameters} + \text{Chronometer } H(z) + \text{BOSS } f\sigma_8\\&\quad\to (H, S, V, R) \text{ defects}\\&\quad\to \mathrm{CCI}_{\mathrm{v0.3}},\ \lambda_a,\ z_c\\&\quad\to \text{figures and JSON.}\end{aligned}$$

12 Data Provenance and Reuse

The Planck parameter values are quoted from the Planck-2018 cosmological parameter analysis [1]. The Cosmic Chronometer values are a compact literature-derived $H(z)$ table compiled from the cited chronometer sources [2, 3, 4, 5, 6, 7, 8]. The growth-connectivity table uses BOSS DR12 consensus $f\sigma_8$ values reported in Ref. [11].

The present work does not claim ownership of the original observational measurements. Reuse of those measurements remains subject to the terms of the original publications, journals, and collaborations. The CSV files and figures in the repository are derived analysis artifacts provided to make this diagnostic benchmark transparent and reproducible. No endorsement by the Planck Collaboration, SDSS/BOSS Collaboration, or the chronometer-data authors is implied.

13 Outlook

The natural next steps are:

- add BAO, DESI, supernovae, weak lensing, and full RSD covariance;
- replace the approximate growth proxy by numerical perturbation evolution;
- compare ΛCDM, w_0w_aCDM, early-dark-energy, and modified-gravity models in the same multi-sector representation;
- test the stability of z_c under alternative datasets and kernel widths;
- extend the scalar CCI into a full MAAT cosmology functional with constraint, anisotropy, and geometric consistency sectors.

14 Conclusion

We have extended the cosmological CCI from a chronometer-only structural-stress projection into a multi-sector benchmark with explicit growth connectivity and robustness margins. The v0.3 implementation defines operational V and R sectors from $f\sigma_8$ and joint expansion–growth consistency, fits companion maximum-entropy sector weights, and derives a reproducible redshift transition proxy from the curvature of the normalised CCI curve.

This does not solve cosmology and does not replace standard inference. It closes a methodological gap in the structural-selection programme: the cosmological CCI now has a concrete path from expansion and growth data to measured sector supports, calibrated weights, and falsifiable structural-stress diagnostics.

References

[1] N. Aghanim et al. (Planck Collaboration), "Planck 2018 results. VI. Cosmological parameters," *Astronomy & Astrophysics* **641**, A6 (2020).

[2] R. Jimenez and A. Loeb, "Constraining cosmological parameters based on relative galaxy ages," *The Astrophysical Journal* **573**, 37–42 (2002).

[3] J. Simon, L. Verde, and R. Jimenez, "Constraints on the redshift dependence of the dark energy potential," *Physical Review D* **71**, 123001 (2005).

[4] D. Stern, R. Jimenez, L. Verde, M. Kamionkowski, and S. A. Stanford, "Cosmic chronometers: constraining the equation of state of dark energy. I: H(z) measurements," *Journal of Cosmology and Astroparticle Physics* **02**, 008 (2010).

[5] M. Moresco et al., "Improved constraints on the expansion rate of the Universe up to $z \sim 1.1$ from the spectroscopic evolution of cosmic chronometers," *Journal of Cosmology and Astroparticle Physics* **08**, 006 (2012).

[6] C. Zhang, H. Zhang, S. Yuan, S. Liu, T.-J. Zhang, and Y.-C. Sun, "Four new observational H(z) data from luminous red galaxies in the Sloan Digital Sky Survey data release seven," *Research in Astronomy and Astrophysics* **14**, 1221–1233 (2014).

[7] M. Moresco et al., "A 6% measurement of the Hubble parameter at $z \simeq 0.45$: direct evidence of the epoch of cosmic re-acceleration," *Journal of Cosmology and Astroparticle Physics* **05**, 014 (2016).

[8] A. L. Ratsimbazafy et al., "Age-dating luminous red galaxies observed with the Southern African Large Telescope," *Monthly Notices of the Royal Astronomical Society* **467**, 3239–3254 (2017).

[9] S. M. Carroll, W. H. Press, and E. L. Turner, "The cosmological constant," *Annual Review of Astronomy and Astrophysics* **30**, 499–542 (1992).

[10] E. V. Linder, "Cosmic growth history and expansion history," *Physical Review D* **72**, 043529 (2005).

[11] S. Alam et al., "The clustering of galaxies in the completed SDSS-III Baryon Oscillation Spectroscopic Survey: cosmological analysis of the DR12 galaxy sample," *Monthly Notices of the Royal Astronomical Society* **470**, 2617–2652 (2017).

[12] C. Krieg, "Cosmological Critical Coherence Index: A Structural-Stress Observable for Cosmic Evolution," preprint and reproducibility bundle (2026).

[13] C. Krieg, *Structural Selection Principle*, ongoing paper series and code repository (2026).